

FFGFT and Vopson's Computational Universe

Parallels and Differences

Johann Pascher

Independent Researcher, Upper Austria, Austria

ORCID: 0009-0000-6518-4064 | Zenodo: 10.5281/zenodo.20117635 (v1.1.0)

This document compares FFGFT with Vopson's Computational Universe hypothesis and Second Law of Infodynamics. Both approaches start from different primitives — FFGFT from T^4 geometry, Vopson from Shannon's information theory — and arrive at structurally very similar conclusions. The convergence is substantial and not coincidental.

Table of Contents

1. Introduction: Two Paths to the Same Destination
 2. Parallel 1: The Minimisation Principle
 3. Parallel 2: Symmetry as Emergent Parsimony
 4. Parallel 3: Gravity as an Emergent Phenomenon
 5. Parallel 4: A Non-Observable Field Explains Dark Matter
 6. Parallel 5: Information and Consciousness as Conserved Quantities
 7. Parallel 6: Information Capacity of the Universe
 8. The Starting Point: Reversed Direction
 9. The Dissent: Is There an Outside?
 10. Torus Resonances as Consciousness Selection Mechanism
 - Summary
 - References
-

1. Introduction: Two Paths to the Same Destination

Melvin Vopson develops the **Second Law of Infodynamics**: information is a physical quantity, information entropy decreases, gravity and symmetry follow from information-theoretic minimisation. [Vopson 2022-2024]

FFGFT begins differently: the conviction that no fundamental contradiction can exist between QM and relativity. The search for the common structure led to T^4 with $\xi = 4/3 \times 10^{-4}$ and $T \cdot m = 1$.

Two independent paths, the same destination. This is not coincidence — it suggests both describe a common deeper reality.

2. Parallel 1: The Minimisation Principle

2.1. Vopson's Second Law of Infodynamics

Information entropy stays the same or decreases (opposite of 2nd law of thermodynamics). Evidence: virus mutations reduce genome entropy; atoms occupy orbitals with lowest information entropy (Hund's rules).

2.2. FFGFT: Resonant Stability

Only integer winding numbers are stable — closed trajectories resonate, open ones decay. The ξ -recursion ladder selects stable levels.

Identical outcome: The system tends toward the geometrically / information-theoretically minimal stable configurations.

3. Parallel 2: Symmetry as Emergent Parsimony

Vopson: Symmetric structures contain less information and save computational power. — **FFGFT:** Symmetric configurations correspond to closed winding trajectories — geometrically minimal, stable fixed points of the ξ -recursion. Symmetry is not coincidence but structural preference.

4. Parallel 3: Gravity as an Emergent Phenomenon

Vopson: Newton's law derived from information minimisation — objects cluster to minimise their shared information footprint.

FFGFT: Gravity from $T \cdot m = 1$ — T-field couples mass to local time structure, gravitational potential from T^4 geometry.

Two independent derivations, the same empirical law.

5. Parallel 4: A Non-Observable Field Explains Dark Matter

Vopson: Dark matter = stored information of the universe. Not directly observable as matter, gravitationally active, structure-bearing.

FFGFT: The T-field: not directly observable as matter, permeates space, gravitational dynamics via $T \cdot m = 1$, carries topological information. No separate

dark matter substance required.

Functionally identical role — different descriptive level.

6. Parallel 5: Information and Consciousness as Conserved Quantities

Vopson: Information cannot be destroyed — consciousness persists in some form. **FFGFT:** Topological invariants (winding numbers) are structurally preserved, even in black holes (Dok. 250). Same conclusion: information does not disappear.

7. Parallel 6: Information Capacity of the Universe

Vopson: $\sim 10^{80}$ – 10^{90} bits (actually occupied information, particles × bits/particle).

FFGFT: Maximum geometric capacity via L_0 :

$$N_{\max}(3D) \approx V_{\text{universe}} / L_0^3 \approx 10^{195} \text{ bits}$$
$$N_{\max}(4D) \approx V_{\{4D\}} / L_0^4 \approx 10^{260} \text{ bits}$$

No contradiction: Vopson counts the contents, FFGFT gives the container size. Fill rate $\sim 10^{-115}$ — the universe uses a vanishingly small fraction of its geometric capacity.

8. The Starting Point: Reversed Direction

	Vopson	FFGFT
Primitive	Shannon information (physical)	T^4 geometry + ξ
Direction	Information → geometry emerges	Geometry → information is projection
Starting point	Landauer, Shannon	QM/RT unification, music theory
Gravity	From information minimisation	From $T \cdot m = 1$
Dark matter	Stored information	T-field from T^4

Both arrive at structurally very similar places because they describe a common deeper reality — approached from different sides. The convergence itself is a signal.

9. The Dissent: Is There an Outside?

9.1. Vopson's Position

If one understood the code, one could theoretically "hack" it — teleportation, overcoming the speed of light, godlike powers. The simulation hypothesis leaves an outside open.

9.2. FFGFT: Structural Exclusion

In FFGFT there is no outside to T^4 . Every "hacking" attempt is itself a process within T^4 . Analogous to Gödel's incompleteness theorem (Dok. 248 §9.4): the system cannot be fully accessed from within, and there is no other vantage point.

The simulation hypothesis presupposes an outside exists. FFGFT structurally excludes this: T^4 is not embedded in a larger space. "What lies outside?" is not an answerable question — analogous to "What lies north of the North Pole?"

10. Torus Resonances as Consciousness Selection Mechanism

Vopson sees the universe as an information-processing system; consciousness is part of it. In FFGFT, torus resonances are a built-in selection mechanism: only integer winding numbers persist. This operates at all scales. Consciousness emerges from the recursive feedback loop (sensor → memory → movement correlation) on the same topological basis.

Summary

Aspect	Vopson	FFGFT
Primitive	Information (physical)	T^4 geometry + ξ
Minimisation	2nd Law of Infodynamics	ξ -resonant stability
Symmetry	Information-minimal	Geometrically minimal
Gravity	From information minimisation	From $T \cdot m = 1$
Dark matter	Stored information	T-field
Info conservation	Yes	Yes (topol. inv.)
Universe bits	10^{80} – 10^{90} (occupied)	10^{195} (max. cap. 3D)
Outside view	Open (simulation possible)	Structurally excluded

Aspect	Vopson	FFGFT
Hackability	Theoretically possible	No — no outside

References

Vopson, M. M. (2022). The mass-energy-information equivalence principle. *AIP Advances*, 9, 095206.

Vopson, M. M. (2023). Second law of infodynamics and its implications for the simulated universe hypothesis. *AIP Advances*, 13, 105308.

Vopson, M. M. (2024). Experimental protocol to test the second law of infodynamics. *AIP Advances*, 14, 035205.

Pascher, J. (2026). *FFGFT v1.1.0*. Zenodo. <https://doi.org/10.5281/zenodo.20117635>

Pascher, J. (2026). *FFGFT — Epistemologische Position* (Dok. 248). Zenodo.

Pascher, J. (2026). *FFGFT — Information und schwarze Löcher* (Dok. 250). Zenodo.